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Assignment 1

Vaviganji Sumanth Babu
192524447

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Q1. Supervised Learning in ML

Definition :

Supervised learning is an ML approach where a model learns a mapping function from input features to output targets using labeled training data. The goal is to accurately predict the target label for unseen inputs.

Key Components :

* Training Data : A set of input-output pairs $D = \{(x_i, y_i)\}$ used to train the model.
* Features (x) : Measurable input attributes describing each data instance.
* Target label (y) : Ground-truth output, categorized as discrete classes for classification or continuous numbers for regression.
* Hypothesis Function (h(x)) : The model function that maps inputs to predicted values.
* Loss Function & Optimization : A metric measuring prediction errors and an algorithm (like gradient descent) that adjusts model parameters to minimize this error.

Example :

* Spam Email Classification :
  - Input : Email text and metadata.
  - Target : Spam or Not Spam.
  - Mechanism : The model learns from labeled emails to classify incoming messages.

Key Issues & Challenges:

* Annotation Cost & Label Noise: Collecting accurately labeled data is expensive and prone to human errors.
* Overfitting & Underfitting: Balancing model complexity so it does not memorize noise or oversimplify trends.
* Class Imbalance: Skewed training data causes the model to favor majority classes.

Q2 Well-Posed Learning Problems

Definition:

According to Tom Mitchell, a computer program learns from experience E with respect to task T and performance measure P, if its performance at task T, as measured by P, improves with experience E.

The Three Defining Components:

1. Task (T): The specific operational activity the system is designed to execute.
2. Performance Measure (P): The metric used to evaluate how well the system performs task T.
3. Training Experience (E): The data, feedback, or practice instances provided to the system.

Why They Are Important:

* Eliminates Ambiguity: Clearly defines the system's scope, inputs, and expected outputs.
* Enables Objective Evaluation: Provides a quantitative benchmark to track progress and prevent subjective testing.
* Guides: The structure of E and T determines the suitable algorithm and learning framework.

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Illustrative Example : Checkers Game

* Task (T): Playing games of checkers.
* Performance Measure (P): Percentage of games won against human tournament players.
* Training Experience (E): Playing practice games against itself.